particularly fascinating, as it follows a bet-hedging strategy, developing two different types of fruit on the same individual with different types of seed dispersal. This plant develops fruit that disperse seeds explosively above ground (Video S1). But it also has non-explosive fruit that develop from floral buds that propel their growth underground (Video S2). As an emerging model system, *C. chenopodiifolia* promises more exciting mechanobiology to come!

Where can I find out more?

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SUPPLEMENTAL INFORMATION

Supplemental information including two videos can be found with this article online at https://doi.org/10.1016/j.cub.2024.07.050.

DECLARATION OF INTERESTS

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Primer

Physics of bacterial chemotaxis

Jeremy P. Moore[1,2] and Thierry Emonet[1,2,3,*]

Chemotaxis is the ability of some organisms to direct their motion in response to chemical signals. This behavior is important across biological scales, from olfactory navigation in animals, to cell migration in development. Chemotaxis is particularly important for bacteria, many of which are capable of directing their motion towards nutrients and away from toxins. The history of bacterial chemotaxis as a field stretches as far back as the late 19th century when scientists observed various species' ability to localize around gas bubbles and pipettes filled with nutrients. After these initial observations, however, it would take over 60 years for the mechanisms of this behavior to come into focus.

During the 1960s, even without modern molecular biology tools, pioneers in the field were able to decipher a surprisingly accurate picture of the internal workings of bacterial chemotaxis. This effort required a host of different experimental techniques, including Julius Adler's classic capillary assays — where bacteria traveled in discrete packets along a capillary tube filled with attractant — and Howard Berg's development of the first microscope capable of real-time, single-cell tracking in three dimensions. However, a critical component in the early success of this field was a healthy appreciation for mathematical modeling and a physics-based approach to solving biological problems. This can be seen in obvious examples like the Keller–Segel model for collective migration (discussed later in this primer) or Berg and Purcell's physics of chemical sensing; but this quantitative streak permeates the entire field, as core concepts from modern systems biology featured in early efforts to discern the mechanism of gradient sensing despite an incomplete picture of the signaling pathway.

Since its inception, the field of bacterial chemotaxis has been highly interdisciplinary. In this primer, we will introduce the problem of chemotaxis starting from the individual bacterium's perspective and ending with the collective migration of bacterial populations. Along with the biology, we will discuss some of the most important mathematical models used to describe chemotaxis at different scales, highlighting both the insights they provide but also their shortcomings. After all, every model is wrong, but the ones described here have been particularly useful in deciphering both the mechanisms and the core logic underlying navigation in bacteria. We have tried to point out the interesting points of failure in these models and what remaining questions they reveal.

The chemotaxis signaling pathway

At its core, the problem of chemotaxis is to choose in which direction to move, given a set of measurements of one's chemical surroundings. Unlike larger eukaryotic cells, bacteria are too small to measure chemical gradients across their cell bodies to make this decision. As such, for bacteria to navigate, they need to move first, and make decisions later. In the case of *Escherichia coli*, having multiple peritrichous flagella (that is, flagella that are distributed around the entire cell) enables only two behaviors: running and tumbling (Figure 1A). In a run, all flagella rotate counterclockwise, propelling the cell forward. If even a single motor switches to the clockwise direction, however, the flagella splay out, causing the cell body to rotate (tumble). After a tumble, the next run will generally be in a different direction from the previous one. With one behavior that generates motion and another that changes the direction of that motion, the specific problem *E. coli* faces is deciding when and for how long to use each behavior.

To make decisions about their behavior, *E. coli* and many other bacteria use a conserved signaling architecture (Figure 1B). Although variations on this theme exist across bacteria, the vast majority of known chemotaxis-signaling pathways share the same general features. Cells express membrane-bound receptor complexes that bind various attractants and repellents and modulate the activity of a kinase, CheA. In *E. coli*, attractants decrease the kinase activity,

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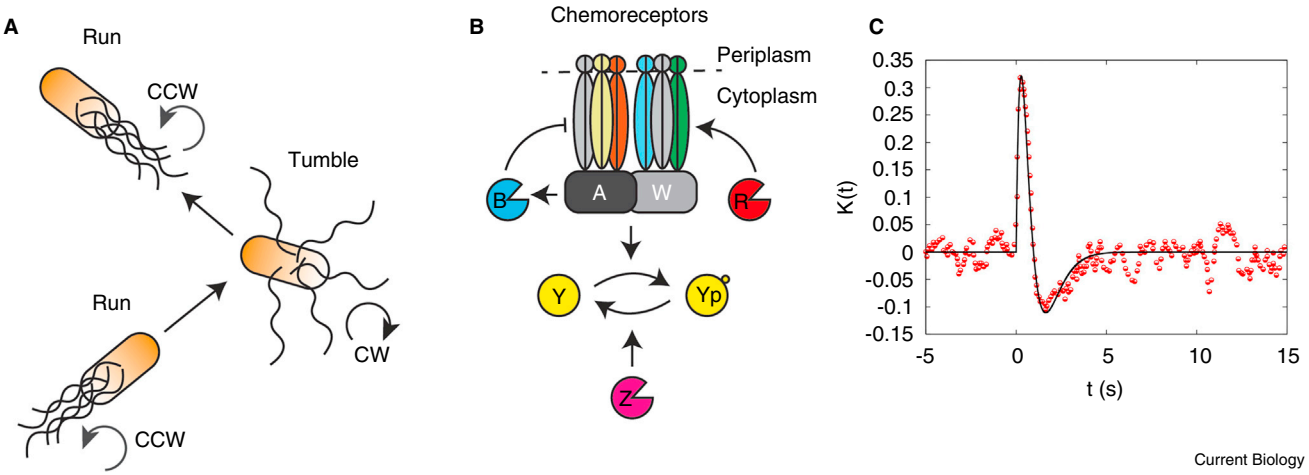


Figure 1. The chemotaxis system of *E. coli*.
(A) *E. coli* navigate by alternating between straight runs and reorienting tumbles. When the flagella rotate counterclockwise (CCW), they form a rotating bundle that propels the cell forward. When they switch to clockwise (CW) rotation, the flagella separate and the cell heading changes. (B) Core components of the chemotaxis signaling pathway. *E. coli* express five chemoreceptor species that form homodimers, which further organize into heterotrimers. These chemoreceptors modulate the activity of a kinase CheA, which clusters with the receptors and a coupling protein CheW. CheA activity is further modulated by methylation and demethylation of the receptors by CheR and CheB, respectively. These methylations oppose changes in CheA activity, keeping it at a constant steady-state level. CheY is the response regulator of the system, and is phosphorylated by CheA and dephosphorylated by CheZ. When phosphorylated, CheY binds the flagellar motor and increases the propensity for clockwise rotation or tumbling. (C) The impulse-response function of the chemotaxis system K(t) is the change in motor bias (probability that the flagella spin counter-clockwise) after a sudden pulse of attractant introduced at t = 0. This response function has two lobes with equal area, typical of systems that adapt precisely, and track the first time-derivative of the signal. (Reused from Celani and Vergassola, 2010, with permission from PNAS, and originally shown by Segall *et al*., 1986.)

whereas repellents increase it. When active, CheA phosphorylates CheY, which binds the flagellar motor. In its phosphorylated state, CheY-P biases the motor in the clockwise direction, leading to more frequent tumbles. Less phosphorylated CheY leads to motors biased in the counterclockwise direction, leading to longer runs.

In addition to the core receptor–kinase complex, there are several ancillary components that are critical for proper signal processing. First, CheY has a dedicated phosphatase, CheZ, the fast action of which ensures that changes in CheA activity lead to almost immediate changes in the steady-state concentration of CheY-P. Second, there are two proteins, CheR and CheB, that respectively add and remove post-translational methylations to chemoreceptors in the kinase-deactivating (off) and kinase-activating (on) conformations. These modifications are added in opposition to changes in the kinase activity. When the kinase activity is low, CheR methylates receptors to increase the activity. When CheA activity is high, CheB demethylates receptors to reduce their activity. CheB can itself be phosphorylated by CheA, speeding up demethylation when kinase activity rises substantially. With these two reactions, if a cell encounters a sudden increase in attractant, the kinase activity, CheY-P, and the probability of a tumble quickly decrease. But then methylation by CheR kicks in, which slowly brings kinase activity, CheY-P, and tumble probability back precisely to their pre-stimulus steady-state levels. This precise adaptation mechanism allows the chemotaxis signaling pathway to sense relative changes in ligand concentration over a wide range.

This ability to adapt to stimuli is essential for navigation. To see why, imagine non-adapting cells swimming in a gradient of a single attractant. If they start at the bottom of the gradient, the concentration is low, so the kinase activity will drop only if they begin swimming up gradient. However, if the cells begin high up on the gradient, the concentration is high in all directions, so the kinase activity will remain low no matter what direction the cells swim. With adaptation, cells are able to dynamically shift their sensitivity so that, over a wide range of ligand concentrations, changes in the concentration can be detected.

Adaptation not only allows cells to detect changes in ligand concentration over a wide dynamic range, it also allows cells to perform interesting computations. Classic experiments measured the rotation of cells attached to a glass slide in response to a sudden pulse of attractant, and quantified counterclockwise bias (the fraction of time the motor rotated counterclockwise) over the total duration of the experiment. With these measurements, it is possible to extract the impulse-response function of the chemotaxis pathway. These response functions have a very interesting shape with two lobes, one above the steady-state bias and one below, both enclosing equal areas (Figure 1C). Bi-lobed response functions like this are typical of systems that compute the first time-derivative of the input signal, and the equal area enclosed in both lobes is due to the precision of adaptation. This response function implies that a bacterium's chemotactic strategy is to compare the recent past with the distant past, using receptor methylation as a 'memory' of previous experiences. By constantly running this comparison, individual cells sense the rate of change in attractant

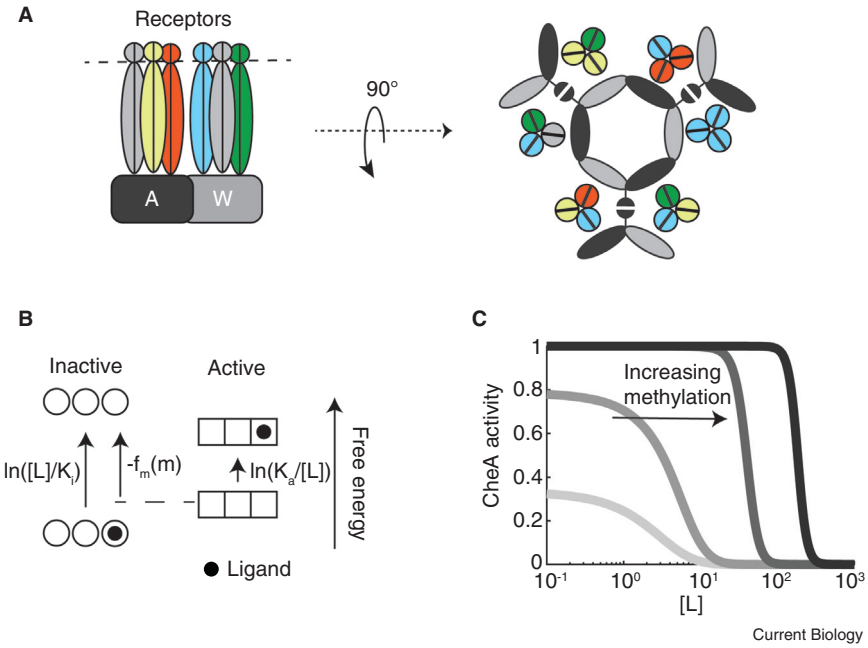


Figure 2. Physics of cooperativity and adaptation in chemoreceptor clusters.
(A) Side view and top-down view of *E. coli*'s chemosensory receptor cluster. The components of the receptor cluster form regular hexagonal lattices. *E. coli* express five different receptor species, which sense a variety of chemical signals including amino acids, dipeptides, sugars, and oxygen. These chemoreceptors can form heterotrimers with each other, and integrate into a regular hexagonal lattice with CheW and CheA dimers. (B) Free-energy diagram of the Monod, Wyman and Changeux model of cooperative receptor clusters. In this framework, the receptor cluster has two states: either all the receptors are in the CheA-inactivating conformation (circles) or in the CheA-activating conformation (squares). Relative free energy of the cluster depends on the current ligand concentration [*L*] and the free-energy contribution from receptor methylation $f_m(m)$, where *m* is the degree of receptor methylation and K_i and K_a are the dissociation constants of the ligand for the receptors in the inactive and active states, respectively. The arrows and text indicate the free-energy change of moving between states. (C) Adaptation through receptor methylation changes the free-energy difference between the on and off states in the absence of ligand. Modulating this difference shifts the center of the dose–response curve.

concentration and modulate their behavior accordingly: lengthening runs if conditions are improving, and increasing tumble frequency if they are worsening.

Signal amplification in chemoreceptor clusters

Now we know what computations cells can perform once they receive a signal. But how does the cell detect signals in the first place? Cells need to amplify very small changes in the concentration of an attractant to large changes in CheA activity. Some biological pathways accomplish this by having a chain of downstream reactions, each of which produces multiple products per input, leading to overall amplification of the signal. In chemotaxis, however, amplification is accomplished in the cluster itself, where a ligand-binding event at one receptor causes that receptor's associated kinase as well as nearby kinases to change their activity. This is thought to be enabled by the structure of the receptor cluster, where receptors, kinases, and coupling proteins are organized in a repeating hexagonal pattern (Figure 2A). Interactions within the receptor cluster allow binding events at only a few receptors to broadly affect kinases within the cluster.

How can we understand this spread of kinase activity throughout the receptor cluster? One description that has been highly successful at predicting experimental results is an allosteric model of cooperative receptors. This view is based on the work of Jacques Monod, Jeffries Wyman, and Jean-Pierre Changeux, who in the 1960s looked for a quantitative description of allosteric transitions in hemoglobin. The Monod–Wyman–Changeux (MWC) model, as it came to be known, takes a unique perspective on the ligand-binding problem. Instead of direct interactions with ligands causing changes in protein conformation, in the MWC model, proteins constantly fluctuate between a high-affinity and low-affinity conformation. At equilibrium, the time spent in each of these conformations is determined by the difference in their free energies. When ligands are present, they preferentially bind and stabilize the higher-affinity conformation, leading to a shift in the equilibrium towards the higher-affinity state.

Things start to get interesting when one considers protein complexes with multiple ligand-binding subunits. Now, the free-energy difference is multiplied by the number of available binding sites. Critically, the MWC framework assumes that receptors switch between conformations in an 'all-or-none' fashion, where all subunits share the same conformation. This leads to a slippery slope of ligand binding: as more ligands bind, the higher affinity state is increasingly stabilized, leading to an even higher average affinity. This increase in average affinity of the 'all-or-none' cluster as more ligands bind leads to steep, cooperative dose–response relationships.

In terms of bacterial chemotaxis, if we take a coarse-grained view of the receptor cluster, we can imagine it as one large allosteric protein complex where each receptor dimer is a binding site, and the entire cluster switches together between kinase-on and kinase-off states (Figure 2B). Thinking about the receptor cluster in the MWC framework gives us an intuitive way to understand both signal amplification and adaptation. Signals are amplified because of strong cooperativity in the receptor cluster. Adaptation through receptor methylation, meanwhile, changes the baseline free-energy difference between the kinase-on and kinase-off conformations in the absence of ligand. As the methylation level increases, the number of ligands that need to bind to achieve a particular kinase activity also increases. As such, increased receptor methylation shifts the dose-response curve of the kinase to higher concentrations (Figure 2C). As ambient ligand concentrations change, the negative feedback of CheR and

CheB on the kinase activity adjusts receptor methylation to keep the dose-response curve centered around the current ligand concentration.

Despite its simplicity, the MWC model accurately predicts the dose-response curves of *E. coli* in a variety of settings. An important experimental technique for measuring these dose-response curves is Förster resonance energy transfer (FRET): the FRET efficiency between fluorophores fused to CheY and CheZ can be used to estimate CheA activity. Using CheY–CheZ FRET, it is possible to directly measure responses of the signaling pathway to changes in ligand concentration. By measuring dose-response curves with FRET, and fitting the MWC model to these curves, the receptor clusters of *E. coli* were revealed to be extremely cooperative.

Beyond measuring cooperativity, CheY–CheZ FRET has also been used to study the speed and accuracy of adaptation. In recent years, single-cell variants of the method have revealed that, in individual bacteria, the activity of the kinase fluctuates dramatically. If one tracks the kinase activity of a single cell over time, it hovers around the steady state, with excursions away from the mean that last for about 10 seconds on average and magnitudes on the order of the average activity. These fluctuations are strongly reduced in the absence of the adaptor enzymes CheR and CheB, suggesting that the futile non-equilibrium methylation–demethylation cycle of the receptors contributes to the fluctuations in kinase activity. However, the magnitude of the fluctuations can also be tuned experimentally in *cheRB* null mutants by lowering the kinase activity with external stimuli, or even by manipulating the relative expression of different receptor species. Expressing only a single receptor can actually lead to massive two-state fluctuations where the cluster alternates between full and no activity with no intermediate states. These experiments suggest that interactions within the receptor cluster itself contribute significantly to fluctuations in kinase activity.

This increased focus on fluctuations in the chemotaxis pathway has brought attention to some shortcomings of the MWC model. Although it agrees well with dose-response experiments, it is limited by being an equilibrium description of the receptor cluster. In the MWC framework, ligands don't cause the cluster to switch conformations; the MWC cluster is constantly fluctuating between two states that cause its subunits to have different ligand affinities. Because of this, if we, for example, increase the cluster's activity by methylating receptors, the cluster spends more time in the lower-affinity active state, so fewer ligands on average will be bound to the receptor cluster. In the MWC framework, a side effect of adaptation is that it forces the cluster to unbind ligands as receptors are methylated. However, we know from multiple different experiments that this is not the case for real chemoreceptors, where average receptor methylation seems to have a very mild effect, if any, on the ligand occupancy.

This discrepancy can be avoided if we drop the requirement from the MWC model that receptor and kinases are in equilibrium with each other. Interest in non-equilibrium mechanisms to describe cooperativity and fluctuations in chemoreceptor clusters has increased dramatically in recent years. The exact nature of cooperativity and adaptation in chemoreceptor clusters and how they are related to fluctuations remain unknown. Unraveling these mechanisms will require close collaboration between theoretical physicists applying non-equilibrium models to the system and experimental biologists considering both the structure and function of the chemoreceptor cluster.

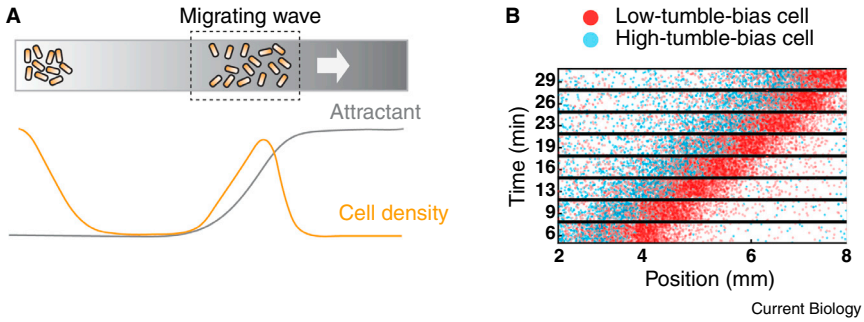


Figure 3. Collective behavior of bacterial populations.
(A) Diagram of the classic capillary assay. Cells are inoculated at one end in a tube filled with liquid medium. Shortly after, waves of bacteria will move at constant speed across the capillary tube. This leads to a steady state where cells are distributed about the attractant gradient in a wave with constant width that moves at constant speed. (B) In these chemotactic waves, cells with different phenotypes can travel together by sorting themselves in space based on their chemotactic capabilities. In liquid medium, cells with high tumble bias trail behind cells with low tumble bias. (Panels A and B reproduced with permission from Fu *et al.*, 2018 (CC BY 4.0).)

Coordinating population-level behaviors in chemotactic waves

In his classic capillary experiments, Adler showed that if you fill a capillary tube with a chemoattractant, bacteria don't just spread evenly throughout the tube. Instead, cells migrate at a constant speed in discrete waves of fixed width (Figure 3A). How does the population stick together when each individual is performing its own random walk? To coordinate their behavior, one might think that cells need some method of communicating. Bacteria are indeed able to secrete various chemoattractants like succinate and serine. Even some signaling molecules involved in quorum sensing, a system of bacterial communication, can be sensed as attractants by the chemotaxis-signaling pathway. However, none of these secretions are essential for the coordinated migration in Adler's experiment. Bacterial populations can coordinate their chemotactic behavior in a more fundamental and elegant way: simply through density-dependent consumption of attractants in the environment.

This interplay between consumption and collective behavior was captured by another classic mathematical model in chemotaxis: Evelyn Keller and Lee Segel's model of traveling *E. coli* bands. Keller and Segel had previously worked on a mathematical framework to understand the aggregation of slime molds, and developed a general model for the chemotactic behavior of single-cell organisms. This general model

was based upon the observation that paths taken by single cells are very similar to Brownian motion, with a slight bias towards higher concentrations of attractant. Such paths can be described by a simple differential equation where a cell's velocity is modeled as the sum of a random diffusive motion and a drift term that accounts for how chemotaxis affects the flow of cells up the attractant gradient. On its own, however, this drift–diffusion process cannot capture the tight coordinated waves in Adler's experiment, instead leading to clouds of cell density that slowly expand over time. To properly capture chemotactic waves, Keller and Segel had to also model the consumption of the attractant over time. By consuming attractant, the population generates a gradient of attractant to follow. Eventually, the population and the attractant gradient reach a steady state where the wave of bacteria and the attractant gradient propagate through space together (Figure 3A).

The Keller–Segel model has seen several extensions and modifications over the years, of which we will highlight only a few. Though insightful, the original version ignored two important effects. The first is that of cell division, which helps the wave reach steady state by compensating for the slow loss of cells that leak at the back of the migrating group and can no longer navigate because the signal is too shallow. In order to keep the wave stable without growth, the original model needed to assume that cells remain sensitive even to infinitesimal changes in attractant concentration. Subsequent studies incorporating growth were able to generate stable attractant waves, but still failed to fit the shape and speed of real migrating populations. It was eventually found that the model had to be modified with a second nutrient that was used for growth, but not chemotaxis. The presence of this second nutrient increases the growth rate of the traveling wave. Then, with more cells present to consume the attractant, the gradient steepens, leading to faster migration.

Another effect ignored by the original Keller–Segel model is phenotypic diversity. Even in isogenic populations, chemotactic abilities vary greatly due to differences in the ability to sense chemoattractants and differences in baseline tumbling frequency. How can populations form a stable traveling wave in the face of such diversity? It turns out that even within the wave, phenotypes can vary dramatically. However, by sorting themselves in space according to their phenotype, it is possible for distinct individuals to travel together at the same speed (Figure 3B). In front of the wave, where the attractant has not yet been consumed, the gradient is shallow. Meanwhile, the gradient is much steeper near the back of the wave, where the population has had a chance to consume some of the attractant. By placing cells with the highest chemotactic ability at the front, and those with the lowest ability at the back, this spontaneous sorting enables each cell to track a gradient matched to its individual capabilities, leading to a mixture of phenotypes that travel at the same speed together. In liquid medium, this means that cells that tumble infrequently tend to lead the pack, whereas cells with high tumble bias trail behind. However, in different environments, phenotypes can be sorted differently. For example, in semi-solid agar, cells with a high tumble bias perform better since their more frequent tumbling allows them to escape from pores in the media. An interesting consequence of this environment-dependent spatial sorting of phenotypic variants is that in different environments, cells with different phenotypes will end up placed at the back of the wave where they are at a higher risk of falling behind. Recent theory and experiments suggest that, when combined with growth, this process results in an adaptation mechanism that adjusts the phenotypic composition of the migrating group to the demands of the environment it traverses, without the need to involve mutations or gene regulation.

Future outlook

This primer has focused mainly on the chemotaxis signaling pathway of *E. coli*. This is partly because it was and remains an important model system for basic physical questions in chemotaxis research. But our focus on this one organism also stems from the fact that far less theoretical attention has been given to other organisms despite decades of extremely interesting experimental work. The chemosensory systems of other bacteria are highly varied, and many are more complex than that of *E. coli*. At the sensor level, there are several components found in other bacteria that *E. coli* don't express. Some bacteria like *Bacillus subtilis* and *Vibrio cholerae* express a poorly understood CheW-like protein, CheV, which has a phospho-accepting domain of unclear function. In addition to different proteins, the chemoreceptor arrays of other bacteria don't always follow the strict repeating pattern of *E. coli*. In *V. cholerae* for example, the expression of CheA is far lower relative to CheW, leading to rings comprised largely of CheW, with rare CheA dimers connecting adjacent rings. The *V. cholerae* chemoreceptor clusters also appear to be far less stable than those found in *E. coli*. Additionally, other bacteria express far more receptor species than *E. coli*, and some even express multiple chemotaxis-like receptor clusters that don't seem to affect chemotaxis at all, but have some yet-unknown function.

At the behavioral level, there is also great diversity in the number and types of flagella that can be expressed. *E. coli*, *Salmonella*, and *B. subtilus* have multiple peritrichous flagella and navigate by running and tumbling. But many species, like members of the *Vibrio*, *Pseudomonas*, and *Aeromondales* families, navigate with a single flagellum, driving the cell either forwards or backwards, depending on the direction of rotation. Still others, like *Borrelia burgdorferi* and *Campylobacter jejuni*, express flagella at both poles that must rotate in opposite directions for the cell to move. Even non-flagellar motility, like twitching, which involves type-4 pili that drag the cell forward, can be controlled by receptor-kinase clusters homologous to the ones that control flagella.

There are many open questions remaining in bacterial chemotaxis. How do flagellar motors interpret CheY-P binding to switch their rotation? How does CheA activity propagate through the receptor cluster? What other processes are controlled by chemotaxis-like receptor clusters? Is there a functional role for fluctuations in the signaling pathway? These are

only a small fraction of the interesting questions that can be answered by close collaboration between the experimental and theoretical sides of the chemotaxis field.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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Primer

Microballistics in fungi and plants

Nicholas P. Money

Ballistic movements in biology are powered by muscle contraction, explosive chemical reactions, the formation and collapse of gas bubbles, merger of fluid droplets, and release of hydrostatic pressure. At the macroscopic end of this kinetic carnival we find jumping fleas, violent spider jaws, shrimp claw hammers, and squirting beetles and cucumbers. The speeds are startling, but the mechanisms seem familiar because they occur on a spatial scale that overlaps with our physical experiences. We jump, albeit more slowly than fleas, for example, and it does not seem strange that seeds will spurt from a swollen cucumber when it hits the ground. Ballistics in microscopic dimensions are very different, operating in a seemingly alien world of fluid mechanics where thin air becomes soup and gravity vanishes.

For very small projectiles, like fungal and plant spores, the only way to escape a surface is to rocket off at an incredible speed at the get go. Even with an initial burst of momentum, many spores travel no farther than a few times their own cell length before viscous drag brings them to a dead stop and they float to the ground. Mushroom spores are discharged in this fashion, using the motion of fluid drops to power their short jumps. Greater discharge distances, ranging from tens of centimeters to meters, are achieved by ejecting spores with pressurized fluid (many fungi), the use of an air gun (*Sphagnum* moss), and a unique snap-through buckling mechanism (the artillery fungus). The effect of air viscosity on motion decreases as the projectile size increases, and gravity assumes more obvious control of the flight path. We return to familiar territory with these longer trajectories whose curvature resembles the arc of a golf ball. To survey these processes, we will use the categories of short-range, intermediate-range, and long-range microballistics.

Short-range microballistics: micrometers to millimeters

Some discharge mechanisms do little more than separate spores from the stalks on which they develop so that they can be dispersed by the surrounding airflow. A fungus called *Deightoniella*, which grows on banana leaves, makes use of explosive bubble formation, or cavitation, within its stalks to launch its spores. Each spore of this fungus sits on a thick-walled bulb at the tip of its stalk, which shrinks in dry air, exerting tension on the cell wall (Figure 1). As drying continues, the water in the cytoplasm fractures to create a vapor-filled bubble, which allows the cell wall of the bulb to pulse outward, shooting the spores at a maximum speed of 0.6 m s^{-1} over a distance of 0.5 mm. The spores travel 15 times their own cell length, which is sufficient for them to escape the boundary layer of still air on the leaf surface. Cavitation relieves the negative hydrostatic pressure

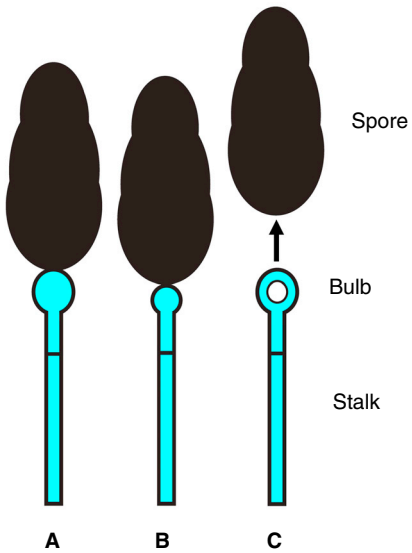


Figure 1. The spore of the fungus *Deightoniella torulosa* is propelled from the bulbous tip of its stalk by the formation of a cavitation bubble.
(A) The spore forms on the tip of its stalk. (B) The bulb at the tip of the stalk shrinks as water evaporates from the cytoplasm, until the cell wall resists further contraction and negative hydrostatic pressure develops within the cytoplasm of the bulb. (C) When the tension within the cytoplasm exceeds a critical threshold, a void or bubble of vapor forms in the bulb, relieving the negative pressure. Bubble formation causes the cell wall of the bulb to expand, which launches the spore.

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